PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Availability, accessibility, and activation of mental health services among university students in Africa: a protocol of a mixed methods systematic review with meta-analysis and meta-synthesis
AUTHORS	Wao, Hesborn; Wao, Melvin A.; Muriithi, Ian; Seraii, Paul; Syonguvi, Jackline

VERSION 1 – REVIEW

REVIEWER	Ferozkhan Jadhakhan University of Birmingham, Institute of Mental Health, School of Psychology, College of Life and Environmental Sciences
REVIEW RETURNED	28-Dec-2021

GENERAL COMMENTS	Overall Impression This is a well done protocol for a systematic review to investigate Availability, accessibility, and utilisation of mental health services among university students in Africa. The authors have followed good research practices to conduct this review and I have a few points for revision/clarification to strengthen the quality of their work. Page 1 line 48: Strengths and Limitations Study heterogeneity may be one concern but high risk of bias, imprecision and low number of studies may be potential factors for limitations. Please amend Abstract: 1. Authors may consider expanding their database searches to include other very useful medical databases, such as Medline and PsycInfo in addition to ERIC. Authors may argue that there is significant overlap between PsycInfo and ERIC, nonetheless it is prudent to include both. 2. Line 36-37: Please review grammar and sentence construction. 3. PROSPERO registration: please check add the CRD reference, also please check reference number it is usually an 11 digit number. Page 2: Introduction: Author may consider exploring rates to a broader geographical area, currently only a few countries are included which may not be truly reflect the current MH prevalence amongst this distinct group. Page 3 Line 14-15 I would argue that CBT and mindfulness are established approaches, author may need to rethink whether a remodelling is

	required or a structured approach to this unique group. Page 3 Line 28 Author may need to include a date when the search will start and end. And also spell out that a rerun of the search will ensue when the full review is completed. Page 3 Line 38 Please add reference(s), I would also consider adding DSM/ICD-10 criteria as the authors mention further in the manuscript. This needs to be spelled out sooner. Page 3 Line 52: Please consider including primary/secondary care to the search strategy. Some studies may explore repeated use of healthcare in distinct areas of healthcare. Page 5 Line 27 Please also consider data provided in figures, how this will be extracted and analysed. Page 5 data extraction line 10. Although the authors suggests that separate data extraction method will be used, it is likely that the data extracted by the two reviewers would be different, therefore clarity needs to be provided how these data will be standardised or an average approach will be conducted or another method employed. Data extraction recommendation: I would suggest that authors extract adjusted and unadjusted data and present them separately. It will strengthen the overall quality of their work. Please consider also to pool outcomes adjusted and unadjusted separately. Page 5, Line 52: This section is not very clear and I would like from the authors to clarify this further. How ORs or RRs will be extracted from continuous outcomes? Please clearly explain in what outcomes (e.g. continuous or binary) the SMD will be used and how the binary outcomes will be analysed. Please add also that adjusted and unadjusted outcomes will be summarized separately Meta-analysis Please explain what model will be run if fixed effect or random effects model is selected (e.g. DerSimonian, Mantel-Haenszel etc) to explore differences between studies, e.g. population, region, assessment methods. If a meta-analysis is appropriate. If not whether the authors will describe their findings narratively. Page 6 line 15: I agree that sources of heterogeneity may arise from individual data characteristics but this is an aggregate meta-analysis and not an individual patient data meta-analysis. Therefore, it will be wiser to conduct a subgroup analysis on study level characteristics to avoid aggregation bias (ecological fallacy). Please explain also if a meta-regression with pre-specified covariates will be performed to identify

	sources of heterogeneity. Suggestion: Authors needs to spell out whether a meta-regression will be conducted, the number of studies required is usually ten. If inadequate a sub-group analysis is advisable which the authors described. This needs to be clarified. Usually is the number is less than ten which analogous to the traditional rule of thumb used to minimize the risk of overfitting in logistic and Cox regression models of at least 10 events per included covariate, a subgroup analysis is advisable. Page 6 line 14: Since the authors intend to test evidence of publication bias, I will suggest that the authors examine publication bias with the "trim and fill analysis" and spell out how imputed missing studies will be dealt with. In the event of presence of publication bias the authors can compare the corrected pooled estimate vs uncorrected. Although the minimum number has been recommended as 10, I think publication bias can be tested with less studies using the trim and fill technique.

REVIEWER	Brhanu Bifftu
REVIEW RETURNED	09-Feb-2022

GENERAL COMMENTS	Better if the background include more detail.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1 (R1) Comments

R2's Comments		**Author's Responses**
1	**Overall Impression**	
1a	This is a well done protocol for a systematic review to investigate Availability, accessibility, and utilization of mental health services among university students in Africa. The authors have followed good research practices to conduct this review and I have a few points for revision/clarification to strengthen the quality of their work.	Thank you.
2	**Page 1 line 48: Strengths and Limitations**	
2a	Study heterogeneity may be one concern but high risk of bias, imprecision and low number of studies may be potential factors for limitations. Please amend	Imprecision and low number of studies meeting inclusion criteria added among limitations
3	**Abstract**	

3a	Authors may consider expanding their database searches to include other very useful medical databases, such as Medline and PsycINFO in addition to ERIC. Authors may argue that there is significant overlap between PsycINFO and ERIC, nonetheless it is prudent to include both	Medline and PsycINFO added among databases to be searched as suggested by R1
3b	Line 36-37: Please review grammar and sentence construction	Grammar corrected.
3c	PROSPERO registration: please check add the CRD reference, also please check reference number it is usually an 11 digit number.	The correct number is now provide: Prospero registration number: CRD42022296870
4	**Page 2: Introduction**	
4a	Author may consider exploring rates to a broader geographical area, currently only a few countries are included which may not be truly reflect the current MH prevalence amongst this distinct group.	Given that the major focus of this paper is on availability, accessibility, and activation of MH services among university students in Africa, not prevalence of MH in this population, we feel that citing a few seminal studies suffice.
5	**Page 3 Line 14-15**	
5a	I would argue that CBT and mindfulness are established approaches, author may need to rethink whether a remodelling is required or a structured approach to this unique group.	We have removed the segment that refers to CBT and mindfulness as we believe its addition does not add value to the prior sentence.
6	**Page 3 Line 28**:	
6a	Author may need to include a date when the search will start and end. And also spell out that a rerun of the search will ensue when the full review is completed	We have included the following statements: *All studies conducted in SSA between January 2000 and February 2022, regardless of the study design, will be included. In addition, a rerun of the search will ensue when the full review is completed.*
7	**Page 3 Line 38**:	
7a	Please add reference(s), I would also consider adding DSM/ICD-10 criteria as the authors mention further in the manuscript. This needs to be spelled out sooner.	We have now added references to the variables defined. However, we left out DSM/ICD-10 as that would mean defining different assessment
8	**Page 3 Line 52**	
8a	Please consider including primary/secondary care to the search strategy. Some studies may explore repeated use of healthcare in distinct	In subsection 2.2 Search strategies, we have included "(primary care) OR (secondary care)" to the search strategy.

	areas of healthcare.	
9	**Page 5 Line 27**	
9a	Please also consider data provided in figures, how this will be extracted and analysed.	In subsection ***2.4.1 Quantitative data extraction***, we have described how we will extract numerical data from figures. How we will analyze such data is explained in the subsection, ***2.6.1 Quantitative data analysis***
10	**Page 5 data extraction line 10**	
10a	Although the authors suggests that separate data extraction method will be used, it is likely that the data extracted by the two reviewers would be different, therefore clarity needs to be provided how these data will be standardised or an average approach will be conducted or another method employed.	In the section, **2.4 Data extraction and management**, the last sentence in the second paragraph reads, "Thus, for each study, data will be independently extracted by two coders and any discrepancies between them discussed until consensus reached or a third reviewer arbitrate." We believe that this sentence addresses the approach proposed in cases where there are disagreement in data extraction.
10b	Data extraction recommendation: I would suggest that authors extract adjusted and unadjusted data and present them separately. It will strengthen the overall quality of their work. Please consider also to pool outcomes adjusted and unadjusted separately.	In subsection ***2.4.1 Quantitative data extraction***, we have added the following sentence regarding data extraction: "To strengthen the quality of our data, we will extract unadjusted and adjusted data and present them separately." How we will analyze such data is explained in the subsection, ***2.6.1 Quantitative data analysis.***
11	**Page 5 Line 52**	
11a	This section is not very clear and I would like from the authors to clarify this further. How ORs or RRs will be extracted from continuous outcomes? Please clearly explain in what outcomes (e.g. continuous or binary) the SMD will be used and how the binary outcomes will be analysed. Please add also that adjusted and unadjusted outcomes will be summarized separately	We apologize for the confusion regarding how ORs or RRs can be extracted from continuous outcomes. In subsection ***2.4.1 Quantitative data extraction***, we have now revised the second sentence to show the type of data we will extract for binary or continuous outcomes. The SMD will be used for continuous outcomes, also explained. In subsection ***2.6.1 Quantitative data extraction***, we have added the following sentence regarding summarizing outcomes:

		"Extracted unadjusted and adjusted outcomes will be pooled and summarized separately."
11b	Meta-analysis: Please explain what model will be run if fixed effect or random effects model is selected (e.g. DerSimonian, Mantel-Haenszel etc) to explore differences between studies, e.g. population, region, assessment methods. If a meta-analysis is appropriate. If not whether the authors will describe their findings narrative	In the subsection, ***2.6.1 Quantitative data analysis***, we have stated that "Meta-analysis will be conducted with a random effects model." In the same paragraph, we have also stated that "Where data are not meta-analyzable, we will describe the findings narratively."
12	**Page 6 Line 15**	
11a	I agree that sources of heterogeneity may arise from individual data characteristics but this is an aggregate meta-analysis and not an individual patient data meta-analysis. Therefore, it will be wiser to conduct a subgroup analysis on study level characteristics to avoid aggregation bias (ecological fallacy). Please explain also if a meta-regression with pre-specified covariates will be performed to identify sources of heterogeneity.	In the 2nd paragraph of section, **2.6 Data analysis and synthesis**, we have added sentences regarding how we will conduct subgroup analysis and the use of meta-regression to identify sources of heterogeneity.
11b	**Suggestion**: Authors needs to spell out whether a meta-regression will be conducted, the number of studies required is usually ten. If inadequate a sub-group analysis is advisable which the authors described. This needs to be clarified. Usually is the number is less than ten which analogous to the traditional rule of thumb used to minimize the risk of overfitting in logistic and Cox regression models of at least 10 events per included covariate, a subgroup analysis is advisable.	As suggested, in section, **2.6 Data analysis and synthesis**, we have added sentences explaining how we will conduct meta-regression, and if not possible, conduct subgroup analysis.
13	**Page 6 Line 14**	
11a	Since the authors intend to test evidence of publication bias, I will suggest that the authors examine publication bias with the "trim and fill analysis" and spell out how imputed missing studies will be dealt with. In the event of presence of publication bias the authors can compare the corrected pooled estimate vs uncorrected. Although the minimum number has been recommended as 10, I think publication bias can be tested with less studies using the trim and fill technique.	In section, **2.6 Data analysis and synthesis**, we have added sentences explaining how we will examine publication bias using funnel plot (with at least 10 studies), or trim and fill technique (for less than 10 studies).

Reviewer 2 (R2) Comments

R2's Comments		Author's Responses
1	**Background**	
1a	Better if the background include more detail.	We have beefed up the Introduction section by including findings from relevant studies focusing on mental health service availability, accessibility, and activation. In doing this, we show that the findings vary, thus providing justification for pooling results together to obtain totality of evidence.

VERSION 2 – REVIEW

REVIEWER	Ferozkhan Jadhakhan University of Birmingham, Institute of Mental Health, School of Psychology, College of Life and Environmental Sciences
REVIEW RETURNED	25-Mar-2022

GENERAL COMMENTS	Well done for the comprehensive response, I am please with your response which will only consolidate this important review.